B.C.S.E. FINAL YEAR EXAMINATION, 2015

(First Semester)

Pattern Recognition & Image Processing

Time : 3 hours | Full marks : 100

Answer question no.1 and any four from the rest

All questions carry equal marks

1. Answer true or false stating reasons $2x10 = 20$

   (a) The success of a pattern classification scheme using decision function depends only on finding the appropriate form of $d(x)$.
   (b) Partitional clustering methods help in exploring data at different levels of granularity.
   (c) Image enhancement techniques are objective, in the sense that they tend to be based on mathematical or probabilistic models of image degradation.
   (d) Data mining engine ideally consists of a set of functional modules.
   (e) The measurement unit used can not affect the clustering of a given data set.
   (f) Each node in a typical ANN can have any number of outgoing connections, where the signals in all of these may not be the same.
   (g) Hopfield network is usually employed for data clustering.
   (h) A data warehouse is usually optimized for validation of incoming data during transactions and it uses validation data tables.
   (i) A typical CBIR system works by using the textual descriptions of a given image.
   (j) Selection of distance metric and integrated region matching criteria are very important in any histogram- based image segmentation method.

2. (a) Describe briefly how pattern classification can be implemented using decision function ? 4

   (b) Assume any three disjoint pattern classes ω_1, ω_2 and ω_3 in $\Re^2$. Compute the decision boundaries and decision rules that are needed to successfully classify all the patterns from the said three classes for the following two cases : 6 + 6

      1. Each class is separable from all other classes
      2. Classes are pair-wise separable

(c) Describe briefly the factors that affect the success of such a pattern classification scheme. 4

3. What is image segmentation? Briefly describe the two basic approaches in segmenting an image. Suppose you do not have any prior information about the number of objects or segments in a given image. Describe a segmentation method that can extract the appropriate segments of the image. 2 + 8 + 10

4. (a) Explain briefly the necessary steps involved in a typical Pattern Classifier. 6

(b) Design a classifier that can classify a set of given text documents into a number of classes depending on their contents where the number of such classes is known *a priori.* Comment on the merits and demerits of such a classifier. 12 + 2

5. What is partitional clustering ? State and explain one such method which is based on the principle of minimization of intra-class distances. Discuss the merits and demerits of the method. 4 + 10 + 6

6. What is content-based image retrieval? What should be the features of such a IR system? Describe how one of those features can be extracted? Describe briefly a supervised image retrieval method for such an image retrieval system. 2 + 2 + 6 + 10

7. Write short notes on any two of the following. 2 X 10

(a) Hyperplane properties
(b) Features of a typical ANN
(c) Typical database vs. data warehouse